Moed B 2015-2016 solution

Question 1

We are looking for a function $G(x, x_0)$ which satisfies:

$$-x^2\frac{\partial^2 G}{\partial x^2} - x\frac{\partial G}{\partial x} + \lambda^2 G = \delta(x - x_0)$$

Note that the operator is not self-adjoint. In order to make it self adjoint, we must divide the equation by x to obtain:

$$-x\frac{\partial^2 G}{\partial x^2} - \frac{\partial G}{\partial x} + \lambda^2\frac{1}{x}G = \frac{1}{x_0}\delta(x - x_0)$$

We can now multiply by x_0 to get:

$$-x_0 x\frac{\partial^2 G}{\partial x^2} - x_0\frac{\partial G}{\partial x} + \lambda^2\frac{x_0}{x}G = \delta(x - x_0)$$

which we can solve as usual.

First we solve the homogeneous equation separately in each of the domains $0 < x < x_0$ and $x_0 < x < 1$. We can work with the original form of the equation for now. Since it is an Euler equation, we expect solutions of the type

$$y \propto x^\alpha$$

plugging this into the equation gives:

$$\begin{aligned} -\alpha(\alpha - 1) - \alpha + \lambda^2 &= 0 \\ \alpha^2 &= \lambda^2 \\ \alpha &= \pm\lambda \end{aligned}$$

therefore

$$G(x, x_0) = \begin{cases} Ax^\lambda + Bx^{-\lambda} & 0 < x < x_0 \\ Cx^\lambda + Dx^{-\lambda} & x_0 < x < 1 \end{cases}$$

We now take into account the boundary conditions:

$$\begin{aligned} |G|_{x=0}| < \infty &\implies B = 0 \\ G|_{x=1} = 0 &\implies D = -C \end{aligned}$$

so

$$G(x, x_0) = \begin{cases} Ax^\lambda & 0 < x < x_0 \\ C\left(x^\lambda - x^{-\lambda}\right) & x_0 < x < 1 \end{cases}$$

We now take into account continuity and jump of the derivative at $x = x_0$. when taking into account the jump in the derivative, we need to use the self-adjoint form, so we have $p(x_0) = -x_0 x$

$$\begin{aligned} G(x, x_0)|_{x=x_0^+} &= G(x, x_0)|_{x=x_0^-} \\ C\left(x_0^\lambda - x_0^{-\lambda}\right) &= Ax_0^\lambda \end{aligned}$$

$$\left.\frac{\partial G(x,x_0)}{\partial x}\right|_{x=x_0^+} - \left.\frac{\partial G(x,x_0)}{\partial x}\right|_{x=x_0^-} = \frac{1}{p_0(x_0)}$$
$$C\left(\lambda x_0^{\lambda-1} + \lambda x_0^{-\lambda-1}\right) - \lambda A x_0^{\lambda-1} = -\frac{1}{x_0^2}$$

these are two linear equations for A and C. The first one gives

$$A = C\frac{x_0^\lambda - x_0^{-\lambda}}{x_0^\lambda}$$

and plugging it into the second we get:

$$\begin{aligned}
C\left(\lambda x_0^{\lambda-1} + \lambda x_0^{-\lambda-1}\right) - \lambda C\frac{x_0^\lambda - x_0^{-\lambda}}{x_0^\lambda}x_0^{\lambda-1} &= -\frac{1}{x_0^2} \\
\lambda C\left(x_0^\lambda + x_0^{-\lambda}\right) - \lambda C\left(x_0^\lambda - x_0^{-\lambda}\right) &= -\frac{1}{x_0} \\
2C x_0^{-\lambda} &= -\frac{1}{\lambda x_0} \\
C &= -\frac{x_0^{\lambda-1}}{2\lambda} \\
A = C\frac{x_0^\lambda - x_0^{-\lambda}}{x_0^\lambda} &= -\frac{x_0^{\lambda-1}}{2\lambda}\frac{x_0^\lambda - x_0^{-\lambda}}{x_0^\lambda} = -\frac{x_0^{\lambda-1} - x_0^{-\lambda-1}}{2\lambda}
\end{aligned}$$

so the solution is:

$$G(x,x_0) = \begin{cases} -\frac{x_0^{\lambda-1} - x_0^{-\lambda-1}}{2\lambda}x^\lambda & 0 < x < x_0 \\ -\frac{x_0^{\lambda-1}}{2\lambda}\left(x^\lambda - x^{-\lambda}\right) & x_0 < x < 1 \end{cases}$$

Question 2

The Lagrangian is

$$L(x,y,y') = y'^2 - y^2 + 4y\cos x$$

The Euler-Lagrange equations:

$$\begin{aligned}
\frac{\partial L}{\partial y} &= \frac{d}{dx}\frac{\partial L}{\partial y'} \\
-2y + 4\cos x &= \frac{d}{dx}(2y') = 2y'' \\
y'' + y &= 2\cos x
\end{aligned}$$

We look for a private solution of the form

$$\begin{aligned}
y^p(x) &= x(A\cos x + B\sin x) \\
(y^p)'(x) &= x(-A\sin x + B\cos x) + (A\cos x + B\sin x) \\
(y^p)''(x) &= x(-A\cos x - B\sin x) + (-A\sin x + B\cos x) + (-A\sin x + B\cos x) = \\
&= x(-A\cos x - B\sin x) + 2(-A\sin x + B\cos x) \\
(y^p)''(x) + y^p(x) &= 2(-A\sin x + B\cos x) = 2\cos x \implies A = 0, B = 1
\end{aligned}$$

So the general solution is:

$$y(x) = x\sin x + C\sin x + D\cos x$$

The boundary condtion $y(0) = 0$ implies $D = 0$. The boundary condtion at $y = \pi/4$ is:

$$\frac{\partial L}{\partial y'}|_{y=\pi/4} = 2y'(\pi/4) = 0$$

for our solution we have

$$y'(x) = \sin x + x\cos x + C\cos x$$

so the boundary condtion gives us

$$\begin{aligned} 0 &= \sin\frac{\pi}{4} + \frac{\pi}{4}\cos\frac{\pi}{4} + C\cos\frac{\pi}{4} = \\ &= \frac{1}{\sqrt{2}} + \frac{\pi}{4}\frac{1}{\sqrt{2}} + C\frac{1}{\sqrt{2}} \\ -1 - \frac{\pi}{4} &= C \end{aligned}$$

so the solution is:

$$\boxed{y(x) = x\sin x - \left(1 + \frac{\pi}{4}\right)\sin x}$$

Question 3

We change to spherical coordinates, so we get:

$$\nabla^2 u(r,\theta,\phi) = 0$$

in the domain $r \geq 1$ with the boundary condition:

$$u(r = 1,\theta,\phi) = \cos\theta$$

Since the problem has cylindrical symmetry, we expect the solution not to depend on ϕ. We no write the Laplacian in spherical coordinates (assuming no ϕdependence):

$$\nabla^2 u(r,\theta) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial u}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial u}{\partial\theta}\right)$$

Guessing a solution

$$u(r,\theta) = f(r)\cos\theta$$

gives:

$$\begin{aligned} \nabla^2 u(r,\theta) &= \cos\theta\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{df}{dr}\right) + \frac{f(r)}{r^2\sin\theta}\frac{d}{d\theta}\left(-\sin^2\theta\right) = \\ &= \cos\theta\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{df}{dr}\right) - 2\frac{f(r)}{r^2}\cos\theta = 0 \end{aligned}$$

This gives an ODE for f:

$$\begin{aligned} \frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{df}{dr}\right) - 2\frac{f(r)}{r^2} &= 0 \\ r^2 f''(r) + 2rf'(r) - 2f(r) &= 0 \end{aligned}$$

This is an Euler equation. Assuming a solution $f(r) \propto r^\alpha$ gives:

$$\alpha(\alpha-1) + 2\alpha - 2 = \alpha^2 + \alpha - 2 = 0 \implies \alpha = -2, \alpha = 1$$

Due to the boundary conditions at $r \to \infty$, we keep only $\alpha = -2$ so we get:

$$u(r,\theta) = A\frac{\cos\theta}{r^2}$$

the boundary condition at $r = 1$ implies that $A = 1$, so we simply get:

$$u(r,\theta) = \frac{\cos\theta}{r^2}$$

which is the potential created by a dipole (this could have been guessed immediately).

Question 4

The solution should have spherical symmetry. We define

$$\eta(r,t) \equiv ru(r,t)$$

and then, as we saw in class, η satisfies the 1D wave equation

$$\frac{\partial^2\eta}{\partial t^2} = c^2\frac{\partial^2\eta}{\partial r^2}$$

on the domain $r \geq 0$ with the boundary condition:

$$\eta(r=0,t) = 0$$

We can define an effective problem on the entire real line $-\infty < r < \infty$ by antisymmetrization of the initial condition. That is, we assume

$$\eta(r, t=0) = r\varphi(|r|)$$

and then since the velocity at $t = 0$ is zero, we get that the solution (using D'Alambert's formula) is:

$$\eta(r,t) = \frac{\eta(r-ct) + \eta(r+ct)}{2} = \frac{(r-ct)\varphi(|r-ct|) + (r+ct)\varphi(|r+ct|)}{2}$$

which gives:

$$u(r,t) = \frac{\eta(r,t)}{r} = \frac{(r-ct)\varphi(|r-ct|) + (r+ct)\varphi(|r+ct|)}{2r}$$

(in fact the absolute value can be removed from $|r+ct|$ as it's always positive).